GESER DUGAROV, PhD

Software Engineer | Distributed Data Systems & AI Data Infrastructure

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profiles: LinkedIn, GitHub

location: Novosibirsk, Russia | Open to relocation

languages: English (advanced), Russian (native)

SUMMARY

Software engineer building petabyte-scale lakehouse and AI data infrastructure at Huawei Cloud, with ownership from technical design through implementation, benchmarking, and production hardening. Author of 50 merged Apache Hudi PRs and three merged Lance PRs; the Flink work reduced end-to-end TPC-H write time by up to 26% and inter-operator data transfer by 29–36%. Current work includes a two-tier distributed cache for Lance/Ray and a query-adaptive, incrementally updatable ANN index targeting tens of billions of vectors. Creator and operator of open-source AI developer tooling, published on PyPI with independent review and verification evidence.

TECHNICAL SKILLS

- **Languages**: Java, Python, SQL; Scala (working knowledge)
- **Distributed data and retrieval**: Apache Hudi, Apache Spark, Apache Flink, Lance, Ray, PostgreSQL, Parquet; lakehouse internals, distributed caching, vector search and ANN
- **Performance and reliability**: Serialization, checkpointing, partitioning/load balancing, profiling, memory/GC optimization, TPC-H benchmarking, monitoring
- **AI developer tooling**: Codex, Claude Code, Antigravity, multi-agent orchestration, task decomposition, human-in-the-loop (HITL) workflows, verification evidence, durable state/recovery, run budgets, trajectory tracing, cost analytics
- **Engineering delivery**: Technical designs/RFCs, automated testing, CI/release engineering, Maven, Poetry, Docker, Linux

WORK EXPERIENCE

May 2023 – Present **Software Engineer / Big Data Engineer**, **Huawei**

Develop core lakehouse, distributed data, and AI retrieval infrastructure for Huawei Cloud; support teammates through code and RFC reviews and technical tutorials.

- Designed and implemented a two-tier distributed cache for Lance vector indices with an in-memory L1, persistent per-worker NVMe L2, and IVF-partition prewarming and routing; integrated it with Ray and benchmarked it on a real cluster.
- Developing a distributed, incrementally updatable, query-adaptive ANN index targeting tens of billions of vectors; authored three merged upstream Lance PRs that bounded IVF-training memory and fixed temporary-resource lifecycle issues, and helped resolve an out-of-memory problem in a three-billion-vector workload.
- Across 50 merged Apache Hudi PRs, authored RFC-84 and implemented its Flink serialization redesign, cutting TPC-H write time by up to 26% and inter-operator data transfer by 29–36%; released in Hudi 1.0.2.

Huawei — continued

- Four additional Flink optimizations ([1], [2], [3], and [4]) improved throughput by 10% and reduced GC overhead by 30%; released in Hudi 1.0.1.
- Diagnosed serial buffer flushing during Flink checkpoints, contributing to a fix that improved checkpoint performance 7×; corrected writer load balancing and timestamp, CDC-merge, and cross-engine record-key defects.
- Simplified setup across 900+ Hudi configuration parameters; designed and implemented partition-level TTL for automated retention and storage-cost control, supported by automated tests and a 100,000-partition load test.
- Drove Hudi's Spark DataSource V2 migration; authored RFC-98 and an initial Copy-on-Write read path to enable additional query pushdowns.

Feb 2022 – May 2023 **Software Engineer / ML Engineer**, **Digital Research** (computer vision startup)

- Designed the event-driven Python backend for a production truck-monitoring system processing ~20,000 images per day; the deployed system reduced fleet idle time by 12%. Also delivered customer reporting/data-visualization and internal monitoring interfaces.

OPEN-SOURCE ENGINEERING

chipping-orchestrator – Creator & Maintainer | Apr 2026 – Present

- Designed and shipped a Python issue-to-PR platform using Codex, Claude Code, and Antigravity, with task decomposition, isolated worktrees, automated verification, independent review, and human-controlled merge. Operated it across 15 repositories in personal use, recording 8,400+ agent runs, including 2,700+ review-agent runs.
- Built GitHub-visible workflow state with host-local coordination, pause/resume, trusted-comment filtering, and interrupted-run recovery; added PR-size limits, lifetime run budgets, human authorization for oversized changes, trajectory tracing, and PostgreSQL/Streamlit cost analytics. Bound reviewer approval to the current PR revision, requirements, developer report, and passing verification evidence.
- Shipped 15 GitHub releases through v0.13.0 (6 Oct 2026), also the first PyPI package; added Poetry packaging, published documentation, and CI on Python 3.12–3.14. As of 13 Oct 2026: 1,068 merged project PRs, OpenSSF Scorecard 8.6/10, and an OpenSSF Best Practices Passing badge.

EDUCATION

PhD, Geophysics, Trofimuk Institute of Petroleum Geology and Geophysics SB RAS

MSc, Computational and Applied Mathematics, Novosibirsk State University